ROS 2 installation and simulator quick start

Developer guide V0.2 | Development preview 1.1.0-preview.1 | 2026-10-11

Luchs A and B development-preview software setup. Start with a software simulator; delivered-hardware support requires separate validation.

Status: Development preview

Choose the software environment

The GitHub development preview contains the shared C++ SDK, ROS 2 driver, messages, launch files and robot descriptions. ROS 1 is in the separate ros1 directory of the same repository. This guide builds only ROS 2.

Environment	Target combination
ROS 2 Jazzy	Ubuntu 24.04; check the version manual for validation status
ROS 2 Humble	Ubuntu 22.04; check the version manual for validation status
ROS 1 Noetic	Separate ros1 workspace; not included in the commands below

Use Linux with the chosen ROS distribution, Git, rosdep and colcon installed. Initialise rosdep if required. This preview is not a formal vehicle-compatibility release.

Get the fixed preview and build ROS 2 only

Use fresh paths for ~/luchs_preview and ~/luchs_ws/src/luchs_ros2; choose different names if those paths already contain your work. Clone the frozen preview release branch, then copy only ros2 into the workspace so colcon does not discover ROS 1 packages.

```
git clone --branch luchs-v1.1.0-preview.1 --depth 1 \
  https://github.com/erfinderwx/rj-platform-developer.git \
  ~/luchs_preview
mkdir -p ~/luchs_ws/src
cp -a ~/luchs_preview/ros2 ~/luchs_ws/src/luchs_ros2
source /opt/ros/jazzy/setup.bash
# Use /opt/ros/humble/setup.bash on Humble.
cd ~/luchs_ws
rosdep install --from-paths src --ignore-src -y
colcon build --symlink-install
source install/setup.bash
```

A successful build should make luchs_bringup and luchs_driver visible to ROS. Source install/setup.bash in every terminal used below. If a package is missing, check the workspace layout and terminal environment.

Start a software simulator first

Choose one command. dry_run:=true selects the built-in simulator instead of physical CAN. No chassis is needed. Keep the adapter disconnected for these examples.

```
ros2 launch luchs_bringup luchs_a.launch.py build:=4ws dry_run:=true
# Or use the Luchs B draft-LCI simulator:
ros2 launch luchs_bringup luchs_b.launch.py protocol:=lci dry_run:=true
```

For another A build, replace 4ws with df, fa or da. In a second sourced terminal, inspect the node and reported state:

```
ros2 node list
ros2 topic echo --once /luchs_driver/state
ros2 topic echo --once /diagnostics
```

Preview boundary. A simulator run demonstrates software behaviour only. It does not establish LCI, IMU feedback, payload switching or other planned functions on a delivered B chassis. The default launch protocol is not evidence of installed firmware.

Read the development manual before commissioning

The version development manual in docs/luchs-development-manual.md records the actual validation results, tests still required and conditions for a formal release. Read it at the same frozen preview release ref as the source.

Validation area	Required evidence before a formal compatibility claim
Linux build and simulator	ROS 2 Humble/Jazzy build, declared dependencies, launch and topic checks
B protocol and parameters	Manufacturer confirmation of interface, geometry, bitrate/IDs, steering sign, current encoding and heartbeat behaviour
Delivered hardware	Matching firmware/harness, measured feedback, control selection, stop and recovery verification on the identified unit

- Match the A build or B interface to the delivered unit; do not treat the software defaults as approved chassis parameters.

- For B, lci refers to the draft interface. Use legacy_b only after VCU V2.6 compatibility is confirmed for that unit; neither selection updates firmware.

- Confirm the adapter, harness and bitrate using the CAN guide and approved unit documentation before connecting.

- Complete the approved commissioning and emergency-stop checks before issuing physical motion commands.

Source and edition

Prepared and reorganised by RJ Tech from the already-public Luchs ROS 2 v1.1.0 package by TerraAgent Robotics, with Git-based setup instructions for the development preview. Source licence: Apache-2.0. The original archive and previous guide edition remain available for provenance.

Original archive source: README.md

Original archive source: luchs_bringup/launch/luchs_a.launch.py

Original archive source: luchs_bringup/launch/luchs_b.launch.py

Development preview ROS 2 source

Version development manual and validation status

Original public software archive

Source licence